

Lecture/Week 2

Absolute and Relative Error

Let x^* be an approximation to an exact quantity x . Then,

The absolute error ' ϵ ' = $|x - x^*|$ and

the relative/percentage error

$$\epsilon_r = \left| \frac{x - x^*}{x} \right| \times 100$$

Example ①

The approximation $\tan x \approx x + \frac{1}{3}x^3$, is valid for small values of ' x ' in radians. Find the relative error in calculation of $\tan x$ at $x = 0.2$

$$\text{Exact value} = X = \tan(0.2) = 0.20271$$

$$\text{Approximate value} = X^* = x + \frac{1}{3}x^3 = 0.2 + \frac{1}{3}(0.2)^3 = 0.202667$$

$$\epsilon_{rel} = \left| \frac{X - X^*}{X} \right| \times 100 = \left| \frac{0.20271 - 0.202667}{0.20271} \right| \times 100$$

$$= 0.0212126 \text{ Ans.}$$

Example (2)

(2)

Two students measure the temperature of the room with the true value of 20°C . Student A measures the temperature as 10°C and the student B measures it as 60°F . Determine the accuracy of these measurements and conclude which student has a better overall accuracy.

Solution

$$\text{Exact value} = X = 20^{\circ}\text{C}$$

$$\text{Approximate value (student A)} = X_A^* = 10^{\circ}\text{C}$$

$$\text{" " (student B)} = X_B^* = 60^{\circ}\text{F} = 15.56^{\circ}\text{C}$$

$$\text{Absolute error (A)} = E_A = |X - X_A^*| = |20 - 10| = 10^{\circ}\text{C}.$$

$$\text{Absolute error (B)} = E_B = |X - X_B^*| = |20 - 15.56| = 4.44^{\circ}\text{C}$$

Student (B) has better accuracy. Ans.

Example ③

③

The width of a rectangular piece of land is measured to be 48.25 ft. If the measurement has a relative error E_r of at most 2%, then what is an upper bound for the absolute error?

Solution

Given $E_r = 2\%$

We know $E_r = \left| \frac{x - x^*}{x} \right| = \frac{|x - x^*|}{|x|}$

$$E_r = \frac{E}{|x|}$$

$$E = \frac{E_r |x|}{100} = \frac{2}{100} \times 48.25$$

$$\boxed{E \leq 0.965}$$